

Place-field repetition is what a predictive map must do

when self-motion cannot resolve the symmetry

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The puzzle

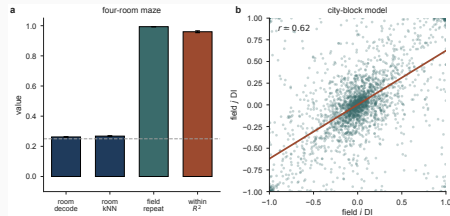
Place cells in identical compartments fire at the same relative location in every one.

“a high degree of spatial repetition”

Spiers et al., 2015

Usual reading: the hippocampus fails to tell the rooms apart.

We show the opposite.



A network trained only to predict its next observation does not represent physical space X .

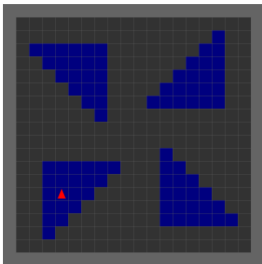
It represents the quotient X/G .

- G is the subgroup that self-motion fails to break.
- Set by equivariance, not by information content.

- **Orbit:** a set of positions related by the symmetry, e.g. $\{x, R^2x\}$ for a 180° rotation.
- **Phase:** which member of the orbit produced the current state.
- **Quotient** X/G : physical space X with every orbit collapsed to one point.
- **Invariant:** unchanged under the group, $\phi(h + g) = \phi(h)$.
- **Orbit-phase accuracy:** can a decoder tell which orbit member produced a given hidden state; chance is $1/|G|$.

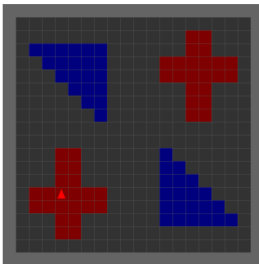
project5_symmetry: new arena layouts

S4: C4 rotational symmetry



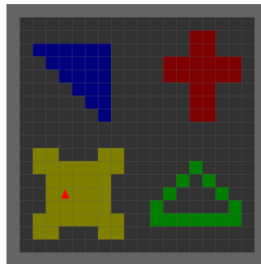
H2 mean=5.54 | landmark tiles=84

S2: 180-degree symmetry



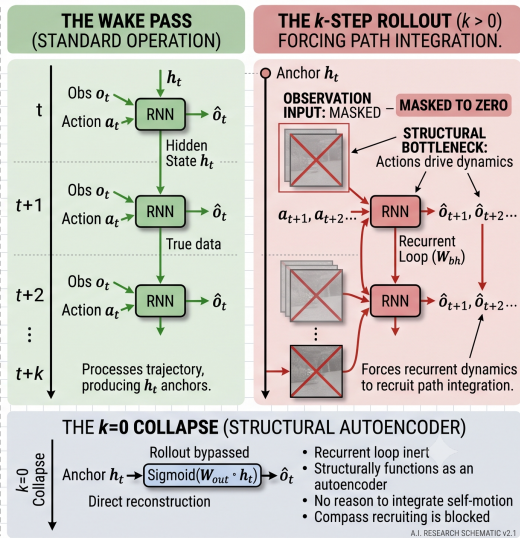
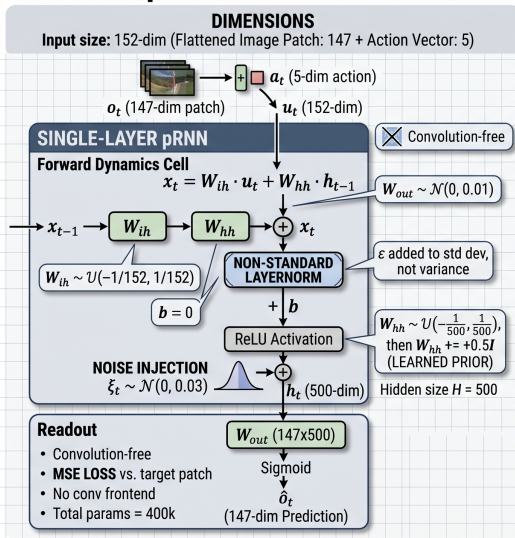
H2 mean=3.21 | landmark tiles=86

S1: asymmetric quadrants

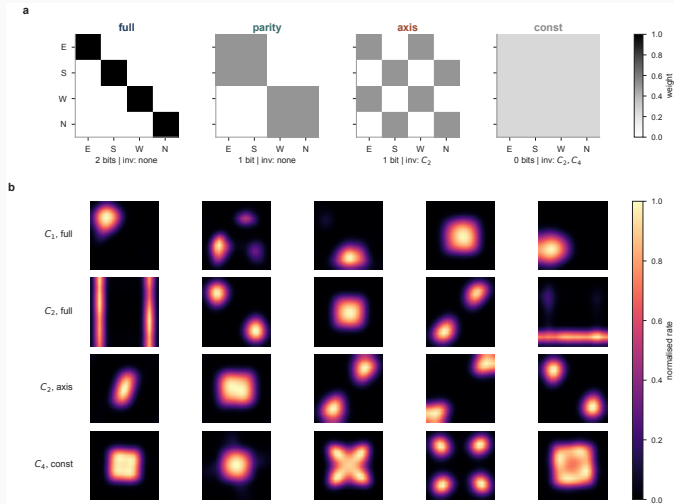


H2 mean=2.15 | landmark tiles=94

pRNN CORE ARCHITECTURE & ROLLOUT DYNAMICS



The manipulation

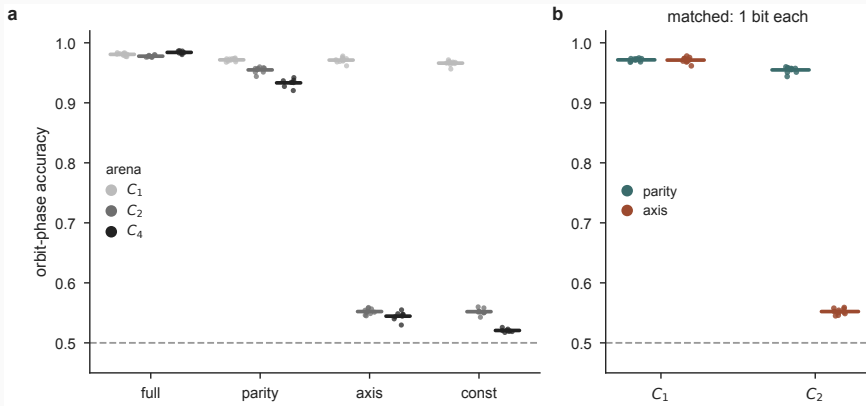


Folding onto the quotient

If the environment has a symmetry G and the head-direction encoding is G -invariant, the posterior over position given any history of observations is G -invariant, and no decoder of the orbit element can beat $1/|G|$.

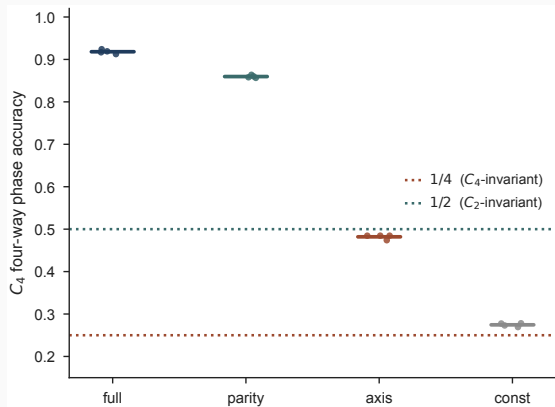
An upper bound, not a prediction. Whether a trained network reaches it is empirical.

Invariance, not information, folds the code



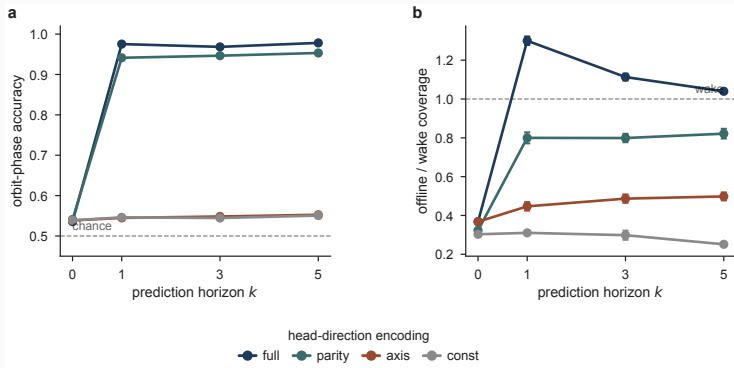
(a) Each dot is one trained network; can a decoder tell which of two 180° -related positions produced this hidden state? *full/parity* stay near 1; *axis/const* collapse to chance, only in the symmetric arenas. (b) *axis* and *parity* carry exactly the same one bit. Indistinguishable in C_1 ; completely separated in C_2 . The only difference between them is which one is invariant to the rotation.

The C_4 ceiling is exactly $1/|G|$



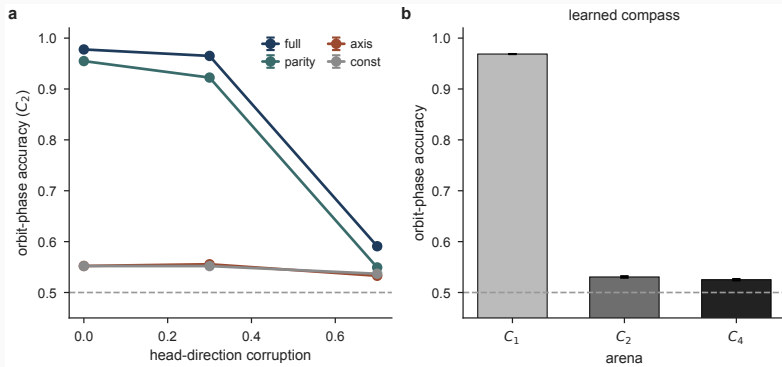
Same test, now decoding among 4 rotational copies (chance $1/4$). *axis* is invariant only under 180° , so it lands on the $1/2$ ceiling; *const* is invariant under the full 90° rotation, so it lands on $1/4$. The dotted ceilings are computed from group theory before seeing the data, not fit to it.

The fold requires prediction



x-axis is the prediction horizon k : how many steps ahead the network is trained to predict. (a) At $k = 0$ (predict the current frame, a pure autoencoder) every encoding sits at chance. At $k \geq 1$ they jump apart and never move again: a step, not a gradient. (b) The same step in offline replay coverage.

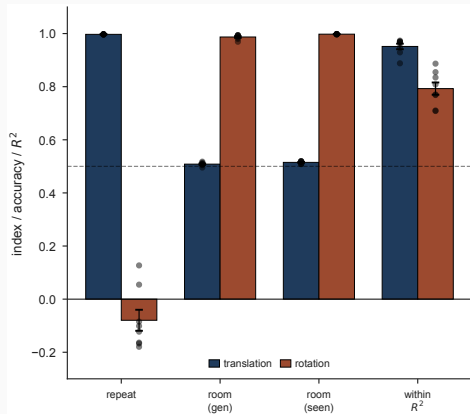
Robust, and unaided by any hand-built encoding



(a) x -axis is what fraction of heading steps are randomly corrupted. *full*/*parity* degrade smoothly as the compass gets noisier; *axis*/*const* stay pinned at chance throughout, since corruption doesn't change whether the signal is invariant. (b) A network given only angular velocity, no absolute heading, still resolves the asymmetric arena and still folds the symmetric one, with no invariant encoding built in by hand.

The same law explains the animal data

Neural



Translated compartments: fields repeat, room undecodable.
Rotated compartments: repetition vanishes, room decodes perfectly. Grieves et al., 2016.

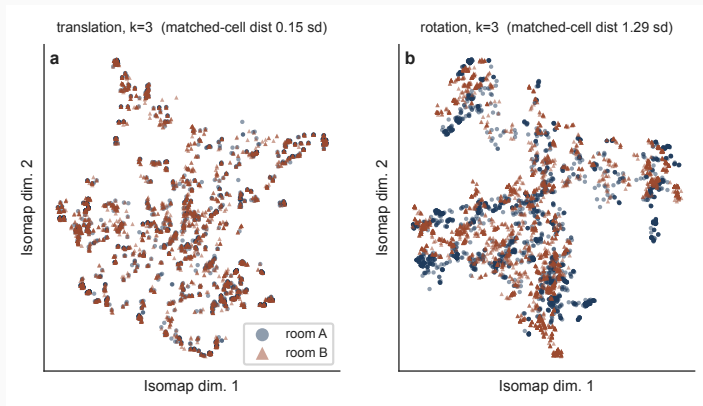
Behavioral

A separate cohort learned to tell the rooms apart.

Learning was significantly impaired exactly where the neural code folds.

“place field repetition constrains spatial learning”

The fold, seen geometrically



Isomap embedding, no decoder involved (Levenstein et al. style). Colour is room identity, never shown to Isomap. Matched-cell distance in the embedding: 0.13 vs. 1.31 (embedding-spread units), $n = 8$ each, $p = 1.6 \times 10^{-4}$.

- The belief process descends to a Markov process on X/G : a MDP homomorphism.
- States that share the same future are bisimilar; this is a bisimulation relation.
- A predictive learner acquiring the successor representation of its belief state should represent that quotient, with no reward and no group given to the network.

- A predictive map represents a quotient X/G , not physical space X .
- Proved as a ceiling, saturated only under a predictive objective.
- One principle explains repetition, neurally and behaviorally.
- One architecture, one real-neuron test that is consistent with the quotient, not proof of it.

A cognitive map represents space only up to the symmetries its own motion cannot break.